

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 2

Duration : 1 Hour
Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I K. Akshitha (19231252) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K. Akshitha

24 + 24 + 23 + 11 + 8
290
BP

Q A 4x4 mini-sudoku grid must be filled with numbers 1-4 such:

Constraints:

Ans:

Given sudoku

let us take the following sudoku with 4 pre-filled cells.

	C_1	C_2	C_3	C_4
R_1	1	-	-	4
R_2	-	4	1	-
R_3	2	-	-	3
R_4	-	1	4	-

Here, '-' represents an empty cell.

Constraints

1. Every row contains 1, 2, 3, 4 without repetition.
2. Every column contains 1, 2, 3, 4 without repetition.
3. Every 2x2 sub-grid contains 1, 2, 3, 4 without repetition.

Step-1:- Assign R_1C_2 .

$R_1 = 1, -, -, 4$, missing numbers are $\{2, 3\}$

Try:

$R_1C_2 = 2$

Then $R_1C_3 = 3$.

	C_1	C_2	C_3	C_4
R_1	1	2	3	4
R_2	-	4	1	-
R_3	2	-	-	3
R_4	-	1	4	-

Constraint checking

Consider the first column:

$1, -, 2, -$

Missing numbers = $\{3, 4\}$.

So R_2C_1 can be 3 or 4.

$$R_2C_1 = 3$$

$$R_2C_4 = 2$$

Now consider the second 2×2 block.

3

4

-

1

now;

	C_1	C_2	C_3	C_4
R_1	1	2	3	4
R_2	3	4	1	2
R_3	2	-	-	3
R_4	-	3	4	-

Step-3:-

$R_3 = 2, -, -, 3$.

Missing numbers = $\{1, 4\}$

• $R_3C_2 = 1$

• $R_3C_3 = 4$.

Step-4:-

$R_4 = -, 3, 4, -$

Missing numbers = $\{1, 2\}$

- $R_4C_1 \rightarrow 4$? No, because now realized $\{1, 2\}$. So check column 1: values are $1, 3, 2, - \rightarrow$ missing 4, causing inconsistency.

This shows that our initial choice $R1C2 = 2$ leads to a contradiction.

Backtracking

At $R1C2$, we had two possible values: 2 or 3
we initially tried:

$R1C2 = 2 \rightarrow$ contradiction.

Therefore, back-track & try:

$R1C2 = 3$.

Now $R1$ becomes: 13 - 4.

Missing number = 2.

So $R1C3 = 2$.

Continue constraint checking.

	C_1	C_2	C_3	C_4
R_1	1	3	2	4
R_2	4	2	1	3
R_3	2	4	3	1
R_4	3	1	4	2

Assignment and Backtracking order

$R1C2 = 2 \rightarrow$ constraint violation detected $\rightarrow$ Backtrack $\rightarrow$
 $R1C2 = 3 \rightarrow R1C3 = 2 \rightarrow R2C1 = 4 \rightarrow R2C4 = 3 \rightarrow$
 $R3C2 = 4 \rightarrow R3C3 = 3 \rightarrow R3C4 = 1 \rightarrow R4C1 = 3 \rightarrow$
 $R4C2 = 1 \rightarrow R4C4 = 2 \rightarrow$ solution found.

Final Answer

1	3	2	4
4	2	1	3
2	4	3	1
3	1	4	2

Thus, Backtracking search systematically assign values.

② Campus Delivery Robot

5x5 Grid

	C_1	C_2	C_3	C_4	C_5
R_1	S	-	P	X	-
R_2	-	X	P	X	-
R_3	-	-	-	P	-
R_4	X	P	X	P	-
R_5	-	-	-	-	G

S = start, G = Hostel/Goal, P = pedestrian zone, - = Normal cell,
X = permanently blocked construction zone.

movement cost

- Normal cell = 1
- Pedestrian cell = 1 + 2 = 3.
- Blocked cell = Cannot be entered.

(a) The constraints directly affect how the robot chooses its path.

1. Four-direction

- up, • down, • left, • Right.

It cannot move diagonally.

2. Blocked Construction Zone.

Cells marked X cannot be traversed.

∴ the robot must avoid these cells & find another safe route.

3. Limited Sensing

The robot can observe only the adjacent cells.

Cannot see the complete 5x5 grid.

∴ it must make decisions using its currently available information.

4. Different movement costs

A normal cell costs 1, while entering a pedestrian zone costs: $1+2=3$

5. Safety requirement

The robot must never enter a blocked construction zone.

(b) Solution using uniform cost search + online search

uniform cost search (UCS) is used because different cells have different costs.

Steps:

1. Start from S.
2. Check the nearby cells and avoid blocked cells.
3. Calculate the cost of each possible move.
4. Choose the path with the lowest total cost.
5. Move next cell and sense the nearby cells again.
6. Update the map.
7. If a new obstacle is found, search for another path.
8. Continue until the robot reaches G.

Example safe path

S → Down → Down → ~~Right~~ → Right → Down → Down
→ Right → Right → G.

(c) AI concepts

Intelligent Agent:

The robot observes, make decisions, moves, and reaches the goal.

Rationality:

The robot chooses a safe path with minimum cost.

Environment:

Property	Type
observability	partially observable
Grid	Discrete
cost	non-uniform
knowledge	Initially unknown
Search	online.

(d) Why UCS + online search?

- UCS finds the minimum-cost path.
- online search works when the robot does not know the complete environment.
- It avoids blocked areas.
- It considers pedestrian-zone costs.
- It updates information while moving.
- It can change the route when new information is found.

Conclusion

UCS + online search is suitable because it helps the robot reach the hostel safely with minimum cost while learning about the environment during movement.